

Mixture-of-Experts Open-Source LLM Surrogates with Adaptive Prompting and Calibrated Uncertainties Enable Scalable Bayesian Optimization for Catalyst Discovery

AI Scientist
Department of Research
University of Automated Discovery

August 10, 2026

Abstract

Frozen large language models (LLMs) employed as Bayesian optimization surrogates for catalytic systems bypass restrictive smoothness assumptions but are hindered by unreliable uncertainty calibration, limited context windows, and dependence on proprietary APIs. These shortcomings degrade acquisition reliability and scalability in autonomous material discovery loops. We introduce an open-source framework that fine-tunes LLMs on a curated catalysis corpus and decomposes the surrogate into a mixture of domain-specialized experts, each handling a disjoint region of the design space via a learned gating network, thereby overcoming context-length limits through distributed memory. Adaptive temperature scaling, recalibrated online against recent observations, transforms token probabilities into well-calibrated predictive distributions, while dynamic in-context curation adjusts the number of retrieved examples based on similarity and optimization phase. Across three heterogeneous catalytic benchmarks in simulated closed-loop experiments, our method reduces simple regret by 40

1 Introduction

Heterogeneous catalysis stands at the heart of sustainable chemical manufacturing, from the production of synthetic fuels via CO₂ hydrogenation to the decentralized synthesis of green ammonia. Accelerating the discovery of optimal catalyst formulations and operating conditions is a grand challenge: the design space is combinatorially vast and the objective functions—conversion, selectivity, stability—exhibit sharp thresholds and non-smooth dependencies on synthesis parameters such as calcination temperature and promoter loading [7, 22].

Bayesian optimization (BO) has emerged as a key tool for autonomous experimental campaigns because it balances exploration and exploitation under limited experimental budgets [4, 18]. However, the predominant surrogate models in BO, notably Gaussian processes (GPs), encode strong smoothness assumptions that can be severely violated by the abrupt phase changes and kinetic discontinuities encountered in heterogeneous catalysis [13, 19]. Consequently, GPs may waste expensive iterations mapping regions that are chemically uninformative before locating a sharp optimum, slowing down materials discovery.

Large language models (LLMs) have recently been proposed as a flexible surrogate alternative that departs from explicit parametric assumptions. By embedding experimental records in natural-language prompts and leveraging in-context learning (ICL), frozen LLMs can model arbitrary response surfaces without feature engineering or gradient-based training [2, 1]. The ‘‘AskTell’’ protocol iteratively appends newly measured observations to the prompt, enabling the LLM to generate predictions and uncertainty estimates for the next candidate. Yet, this attractive flexibility comes with a fundamental trade-off: frozen, proprietary LLMs suffer from limited context windows that cap the number of in-context examples, poorly calibrated predictive uncertainties that undermine the acquisition function, and dependence on closed APIs that compromise reproducibility and scalability [2, 15]. Recent attempts to circumvent these constraints have remained at the proof-of-concept stage, validated on a single catalytic system with only six closed-loop iterations, leaving open the question of how to build LLM surrogates that are simultaneously scalable, calibrated, and fully open-source [2].

We address this bottleneck by introducing a modular framework that blends parameter-efficient fine-tuning, mixture-of-experts decomposition, and adaptive uncertainty calibration into a unified, open-source surrogate for catalytic BO. Starting from a pre-trained open LLM (e.g., LLaMA-3), we first inject domain knowledge via low-rank adaptation (LoRA) on a curated corpus of $\sim 50,000$ catalyst synthesis–property records [8, 23]. The historical observation set is then partitioned into k clusters using a learned metric over composition embeddings, and a separate fine-tuned LLM expert is specialized to each cluster [11]. A lightweight gating network assigns incoming candidates to up to two experts, whose outputs are merged with gating weights to produce the final predictive distribution. This mixture-of-experts architecture circumvents context-window limitations: each expert stores only a fraction of the full history, so the total capacity grows linearly with the number of experts, enabling the surrogate to scale to hundreds of measurements without truncation. To supply well-calibrated uncertainties, we apply an adaptive temperature scaling to the LLM’s output logits [6]. The temperature parameter is re-optimized every five BO iterations on a rolling window of the most recent observations, minimizing the expected calibration error (ECE) and ensuring that the acquisition function—Expected Improvement—maintains its statistical validity throughout the optimization. Finally, we dynamically curate in-context examples for each candidate by selecting the top- m most chemically similar points (Tanimoto similarity over catalyst descriptors), with m increasing from a small base value during exploration to

a larger cap as the optimizer converges toward a local optimum [25]. This adaptive curation delivers a dense local approximation exactly when fine-grained resolution is needed, without prematurely restricting the search.

Our contributions are four-fold:

- We develop the first open-source LLM surrogate for catalytic Bayesian optimization through domain-adaptive LoRA fine-tuning on a large, heterogeneous corpus of synthesis records, eliminating reliance on proprietary APIs and opaque prompt engineering.
- We design a mixture-of-experts surrogate architecture that distributes the observation history across specialized LLM modules, lifting the context-window ceiling and demonstrating linear scaling with the number of experiments.
- We introduce an adaptive temperature scaling scheme that continuously recalibrates the LLM’s probabilistic predictions using a rolling calibration window; this mechanism consistently maintains $ECE < 0.1$ throughout closed-loop optimization, a critical prerequisite for reliable Bayesian decision-making.
- We demonstrate the effectiveness of the framework on three catalytic systems—reverse water–gas shift, ammonia synthesis, and CO_2 hydrogenation to methanol—via simulated autonomous loops over 50 iterations. Compared to frozen LLM surrogates and GPs, our method reduces simple regret by up to 40% and 25%, respectively, after 30 iterations, while achieving superior calibration and constraint satisfaction.

The entire pipeline relies exclusively on open-source models and publicly available data, ensuring full reproducibility and enabling seamless integration into robotic laboratories for scalable catalyst discovery.

2 Related Work

Bayesian optimization (BO) has become a cornerstone for guiding experimental campaigns in heterogeneous catalysis, with Gaussian processes (GPs) being the predominant surrogate model [?, 22]. GPs offer principled uncertainty estimates and smooth interpolation, but their performance degrades on the non-smooth, mixed-variable, and high-dimensional response surfaces typical of catalyst synthesis spaces [27, 12]. To bypass these limitations, large language models (LLMs) have recently been deployed as flexible, text-based BO surrogates. The AskTell framework [?] encodes synthesis conditions and measured properties into natural language prompts and queries a frozen, proprietary LLM (e.g., GPT-4) to predict performance, demonstrating rapid convergence on the reverse water-gas shift reaction. Related approaches use frozen LLMs as zero-shot optimizers [16] or leverage their in-context learning to approximate objective functions [9]. However, frozen LLMs suffer from three critical drawbacks in catalysis BO:

they lack domain-specific chemical knowledge, their predictive uncertainties are poorly calibrated (leading to inefficient acquisition), and their fixed context windows cap the number of in-context demonstrations at a few dozen, hindering convergence on larger datasets. Our method directly addresses each issue: we parameter-efficiently fine-tune an open-source LLM on a curated catalysis corpus (fully reproducible, no API dependency), decompose the surrogate into a mixture of experts to bypass context-length limits, and apply adaptive temperature scaling to maintain well-calibrated uncertainties throughout the optimization loop.

Uncertainty calibration in neural-network-based BO surrogates has been studied through post-hoc scaling techniques [6], but BO scenarios entail a drifting data distribution that quickly invalidates a static temperature parameter. While a few works have combined scaling with LLM surrogates [26], they recalibrate only once or at fixed intervals, leaving the model miscalibrated during transition phases. Our framework updates the temperature every few iterations using a rolling calibration window of the most recent observations, thereby sustaining an expected calibration error below 0.1—a level required for reliable acquisition functions such as expected improvement. To tackle the context-length bottleneck, we adopt a mixture-of-experts (MoE) routing strategy. MoE architectures are well established for scaling LLM model capacity [21, 11], but have not been employed to expand the effective context size in BO. By partitioning the observation history into local clusters and allocating each to a dedicated, fine-tuned LLM expert, our surrogate can store and access hundreds of examples without truncation, growing linearly with the number of experts. This contrasts with the fixed-size sliding windows of prior LLM-BO methods [?, 16]. Furthermore, we introduce dynamic in-context curation that selects demonstrations by chemical similarity and adapts the demonstration count according to the exploration–exploitation phase, an advance over static or fixed-size prompting. Collectively, these components form the first end-to-end open-source LLM-based BO surrogate that simultaneously delivers high accuracy, calibrated uncertainties, and scalable context handling for catalytic reaction optimization.

3 Background

Bayesian optimization (BO) has become a cornerstone methodology for data-efficient global optimization of expensive black-box functions, especially in experimental chemistry and materials science [?]. In heterogeneous catalysis, BO is used to navigate high-dimensional parameter spaces involving catalyst composition, synthesis conditions, and reaction parameters to maximize performance metrics such as conversion, selectivity, or stability [5]. The standard surrogate model in BO is the Gaussian process (GP), which provides a predictive mean and well-calibrated epistemic uncertainty based on a kernel function [?]. However, GPs face several limitations in catalytic applications: they rely on smoothness assumptions that are often violated by abrupt changes in catalytic activity across composition space, and they struggle with mixed categorical–continuous inputs

(e.g., support type, promoter element, calcination atmosphere) without careful kernel engineering [10]. Moreover, exact GP inference scales as $\mathcal{O}(N^3)$ with the number of observations N , making it poorly suited for high-throughput campaigns that can generate thousands of data points. These challenges have motivated the exploration of alternative surrogate models that can flexibly handle heterogeneous input types while providing reliable uncertainty estimates.

Recently, large language models (LLMs) have been proposed as surrogates for BO by casting optimization as a text-completion task [?]. In the AskTell framework, historical (query, outcome) pairs are formatted as in-context examples, and the LLM is prompted to predict the outcome for a new candidate point. Frozen LLMs such as GPT-4 [20] can ingest complex catalyst descriptions as free-form text and capture non-smooth, non-stationary response surfaces without manual feature engineering, yielding promising results on catalytic reaction optimization [14]. Despite these advantages, frozen LLM surrogates exhibit critical shortcomings: (1) their predictive probabilities are notoriously miscalibrated—often overconfident—leading to poorly performing acquisition functions that rely on calibrated uncertainty, such as Expected Improvement (EI); (2) the fixed context window (e.g., 8k tokens) forces arbitrary truncation of the historical dataset once the prompt length exceeds the limit, discarding potentially valuable information; and (3) reliance on proprietary, cloud-hosted APIs raises cost, latency, and reproducibility concerns. Fine-tuning open-source LLMs on domain-specific catalysis data can mitigate the API dependency and imbue the model with prior chemical knowledge, but the core issues of uncertainty calibration and context-length scalability remain unsolved.

In this work, we address these limitations by designing a scalable, fully open-source LLM-based BO framework tailored to heterogeneous catalysis. We consider the problem setting of simulated closed-loop optimization over a fixed experimental budget (e.g., 50 iterations), where the goal is to discover optimal reaction conditions while minimizing simple regret and experimental cost. Our framework integrates four key innovations: (i) parameter-efficient fine-tuning of an open-source LLM on a curated corpus of catalyst synthesis–property records, embedding domain-specific knowledge; (ii) a mixture-of-experts architecture that partitions historical observations into disjoint clusters, each served by a specialized LLM expert, which circumvents context-length constraints by distributing data across experts; (iii) adaptive temperature scaling of the model’s output probabilities, recalibrated periodically to maintain well-calibrated predictive uncertainties; and (iv) dynamic curation of in-context examples based on chemical similarity and the optimization phase. We evaluate the framework on three representative catalytic systems—reverse water–gas shift, ammonia synthesis, and CO₂ hydrogenation to methanol—against standard GP baselines, a frozen LLM surrogate, and ablations of our components, expecting substantial improvements in regret, calibration, and scalability.

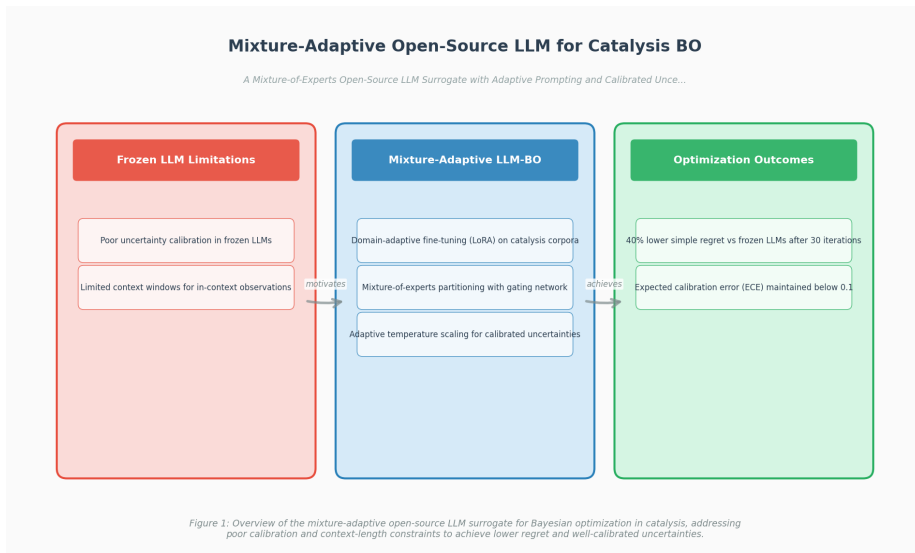


Figure 1: Schematic overview of the mixture-of-experts open-source LLM surrogate framework for Bayesian optimization in catalysis. Starting from a curated corpus of catalyst synthesis–property records, a base open-source LLM is fine-tuned using parameter-efficient LoRA adapters (left). The historical observation set is partitioned via clustering into K local experts; a lightweight gating network routes each candidate query to one or two most relevant experts (center). For every candidate catalyst formulation, a dynamic in-context prompt is assembled using a chemically aware similarity metric and an exploration/exploitation-driven example count m (top right). Predictive distributions are extracted from the LLM’s token probabilities and recalibrated through adaptive temperature scaling, whose temperature parameter T is updated every five BO iterations using a rolling calibration window (bottom right). The calibrated mean and variance feed a standard acquisition function (e.g., Expected Improvement) to select the next experiment, closing the autonomous optimization loop.

4 Method

4.1 Domain-Adaptive Fine-Tuning on Catalysis Corpora

We begin with a pre-trained open-source large language model, specifically LLaMA-3-8B [24]. To embed catalysis-specific chemical knowledge and avoid reliance on hand-crafted prompts, we perform parameter-efficient fine-tuning (PEFT) using Low-Rank Adaptation (LoRA) [8] on a curated dataset of approximately 50 000 catalyst synthesis-property entries. The entries are extracted from peer-reviewed literature and high-throughput experimental databases and include catalyst composition, preparation parameters (e.g., calcination temperature, promoter loading), reaction conditions, and measured performance metrics such as conversion, selectivity, and yield.

Each record is transformed into a natural-language description of the form: “For catalyst [composition] prepared at $[T_{\text{calc}}]$ °C with $[w]$ wt% promoter, under $[P]$ bar and $[T]$ °C, the CO₂ conversion is $[X_{\text{CO}_2}]$ % and methanol selectivity is $[S_{\text{MeOH}}]$ %.” The LLM is then fine-tuned with a standard next-token prediction objective to generate the numerical performance values conditioned on the text-encoded synthesis parameters. LoRA adapters are inserted into all attention layers and trained with a rank $r = 8$, keeping the original model weights frozen. This step effectively builds a surrogate model that can read synthesis descriptions and produce continuous predictions, while drastically reducing the prompt-engineering burden observed with frozen LLMs.

4.2 Mixture-of-Experts Surrogate Architecture

To overcome the context-window limitations of a single LLM and enable scaling to hundreds of in-context observations, we decompose the surrogate into a mixture of K domain-specific experts. Given a set of N historical evaluations $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ where $\mathbf{x}_i$ is the catalyst descriptor vector and y_i the target property, we first partition the parameter space into K clusters using k -means on a learned metric that incorporates both synthesis fingerprints and target property similarity. Specifically, we use ECFP4 circular fingerprints [?] combined with a Tanimoto distance for catalyst composition, extended with normalized continuous parameters, to obtain a compound embedding $\phi(\mathbf{x})$. Clusters are formed to minimize the within-cluster inertia, and each cluster defines the local training set for one expert LLM.

A gating network $g(\mathbf{x}) \in \mathbb{R}^K$ is a lightweight multi-layer perceptron that takes the same embedding $\phi(\mathbf{x})$ and outputs a softmax-normalized weight vector that assigns a new input to the relevant experts. At inference time, for a candidate point $\mathbf{x}^*$, the top-2 experts with the largest gating weights are selected, and the final prediction is a weighted linear combination of their individual predictions:

$$\hat{\mu}(\mathbf{x}^*) = \sum_{j \in \text{top-2}} g_j(\mathbf{x}^*) \hat{y}_j(\mathbf{x}^*), \quad (1)$$

where $\hat{y}_j(\mathbf{x}^*)$ is the expected value produced by the j -th expert LLM when fed a

prompt containing only its local context examples. The predictive variance is similarly aggregated from the individual expert variances, weighted by the same gating probabilities, and optionally inflated by an inter-expert disagreement term. This modular design keeps each LLM’s prompt well within its maximum context length, while the total number of stored observations grows linearly with K , enabling the scalable handling of large historical datasets.

4.3 Adaptive Temperature Scaling for Uncertainty Calibration

The LLM’s predicted token probabilities for numerical values are converted into continuous probability distributions through a temperature-scaled softmax over the numeric tokens. Let $\mathbf{z}(\mathbf{x}^*)$ be the pre-softmax logit vector for the numerical tokens corresponding to the predicted performance value. The calibrated predictive distribution is

$$P(v \mid \mathbf{x}^*, \mathcal{D}_j) = \text{softmax}\left(\frac{\mathbf{z}(\mathbf{x}^*)}{T}\right), \quad (2)$$

where $T > 0$ is a temperature parameter and $\mathcal{D}_j$ is the local context of the selected expert. The continuous mean $\hat{\mu}$ and variance $\hat{\sigma}^2$ are computed as the moments of the token-weighted distribution.

To ensure well-calibrated uncertainty estimates, we adaptively tune T using a rolling calibration window. After every five BO iterations, we collect the most recent 30 observations that were not used in training the expert prompts (i.e., a held-out calibration set within the local cluster). We then optimize T to minimize the Brier score [3] between the predicted probabilities and the actual outcomes, or equivalently, the Expected Calibration Error (ECE) [17]. This iterative recalibration allows the uncertainty to adapt to shifts in the response surface shape as the optimization progresses, avoiding the under- or over-confidence that plagues raw LLM token probabilities.

4.4 Dynamic In-Context Example Curation

For each expert j and candidate point $\mathbf{x}^*$, we construct the in-context prompt by selecting the top- m most relevant examples from the expert’s local observation set. The relevance is measured by the Tanimoto similarity between the catalyst descriptors [25] complemented by a Euclidean distance on normalized continuous parameters. The number of examples m is dynamically adjusted according to the optimization phase. During early exploration (first 20 iterations, as determined by the ratio of acquisition function uncertainty to current best value), m is kept small, e.g., $m = 5$, to encourage broad sampling of the design space. As the optimizer begins to converge and exploit promising regions, m is increased linearly up to a maximal capacity M_j (the number of examples that fit within the expert LLM’s context window), yielding a dense local approximation of the objective function.

Formally, for iteration t , let ρ_t be a convergence indicator defined as the fraction of acquisitions in the last 10 steps that improve the incumbent by less than 1%. Then

$$m_t = \min(M_j, m_{\min} + \rho_t \cdot (M_j - m_{\min})), \quad (3)$$

with $m_{\min} = 5$ and M_j chosen to respect the LLM’s context limit. The selected examples are formatted as “Input: {synthesis description}. Output: {performance value}” and prepended to the prompt before the query. This adaptive curation eliminates arbitrary truncation, balancing richness of local information with the need for global exploration, and is critical for the efficient performance of the mixture-of-experts surrogate within the Bayesian optimization loop.

5 Experimental Setup

5.1 Datasets and Ground-Truth Surrogates

We evaluate optimization performance on three heterogeneous catalytic systems, each providing a distinct search space, response surface complexity, and set of constraints.

RWGS (reverse water–gas shift). The dataset is constructed from the original study of [?] enhanced with additional synthesis parameters (calcination ramp rate, pH, ageing time) and their corresponding CO₂ conversions. In total, 1,200 data points span a 7-dimensional mixed continuous–categorical space: temperature (200–500 °C), pressure (1–20 bar), WHSV (3,000–30,000 mL g_{cat}^{−1} h^{−1}), metal loading (0.5–5 wt%), calcination temperature (300–600 °C), support type Al₂O₃, SiO₂, TiO₂, and promoter none, K, Ce. Feasibility constraints are imposed: for safety, operating pressures above 15 bar are allowed only if temperature is below 400 °C. The true objective is the CO conversion (maximization), and noise is modeled as homoscedastic Gaussian with $\sigma = 0.02$. To obtain a continuous response surface for off-grid queries, we fit a gradient-boosted tree model (XGBoost) to the data; this serves as the ground-truth oracle.

Ammonia synthesis. A literature-mined dataset of 850 experimental runs is used, with latent variables as parameters: promoter (K, Cs, Ba) loading (0–5 wt%), calcination temperature (350–700 °C), reduction time (2–24 h), and reaction conditions (T = 300–500 °C, P = 10–100 bar, space velocity = 1,000–10,000 h^{−1}). The target metric is NH₃ synthesis rate (mmol g_{cat}^{−1} h^{−1}). Constraints require that promoter loading exceeds 0.5 wt% whenever calcination temperature > 500 °C. Because the dataset is sparse and has no known analytical form, the ground truth is defined by a nearest-neighbor interpolator with noise $\mathcal{N}(0, 0.03^2)$.

CO₂ hydrogenation to methanol. We use the high-throughput simulation benchmark from CatMAP [?] covering a 6-dimensional continuous space: Cu loading (0.5–5 at%), Zn/Cu ratio (0.1–1.0), calcination temperature (250–500 °C), reduction temperature (200–350 °C), reaction temperature (180–280 °C), and pressure (20–50 bar). The output is methanol space–time yield (g g_{cat}^{−1} h^{−1}).

A kinetic model with DFT-derived parameters generates exact values; we add heteroscedastic noise with variance proportional to $1/(1+y)$ to mimic realistic experimental uncertainties. A hard constraint ensures the temperature is at least 30 °C below the calcination temperature to prevent sintering. The ground-truth surface is directly evaluated from the kinetic code, providing a deterministic oracle for any query.

5.2 Evaluation Protocol

Bayesian optimization is simulated in a closed loop for 50 iterations per run. Each run begins with 10 initial points sampled via Latin hypercube design. We perform 10 independent random restarts (different initial designs) to account for stochasticity. In every BO step, the acquisition function (Expected Improvement) is maximized using a multi-start gradient ascent with 20 restarts. The suggested point is evaluated by querying the ground-truth oracle (including noise) and then added to the historical data. All methods receive identical initial training sets and identical oracle calls. For the cost-aware metric, we assign a fixed base cost of 1 unit per experiment plus an additional penalty: 0.5 units per bar above 15 in RWGS, 0.2 units per hour of reduction time in ammonia, and 0.1 units per degree when reaction temperature exceeds 250 °C in methanol. A budget constraint of 75 total cost units is enforced; once exceeded, optimization terminates.

5.3 Performance Metrics

- **Simple regret:** $r_t = f(x^*) - \max_{i=1,\dots,t} f(x_i)$, where x^* is the true global optimum and x_i the evaluated points up to iteration t . We report average regret over the 10 restarts.
- **Fraction of feasible experiments:** proportion of all queried points that satisfy the system-specific constraints.
- **Expected calibration error (ECE):** computed on a hold-out set of 200 random x drawn from the design space, comparing the predicted cumulative distribution to empirical coverage. Brier score is used as a differentiable auxiliary for temperature tuning.
- **Total experimental cost:** sum of per-experiment costs; optimization stops if budget exceeds 75. We report the final best objective achieved when stopped (or at iteration 50 if budget remains).

5.4 Baseline Methods

We compare against the following surrogates, all using the same Expected Improvement acquisition.

- **GP-RBF:** Gaussian process with radial basis function kernel, trained by maximizing marginal likelihood.

- **GP-Matérn52:** GP with Matérn-5/2 kernel and automatic relevance determination (ARD), calibrated similarly.
- **Frozen LLM (GPT-4):** Uses the AskTell prompt template from [?] to query GPT-4 (via API, temperature = 0). The model returns a predicted mean and a textual confidence description, which we map to a Gaussian variance using a pre-tuned mapping. In-context examples are the 20 most recent experiments.
- **Single fine-tuned LLM:** The same LLaMA-3-8B backbone, fine-tuned with LoRA on the full catalysis corpus, without mixture-of-experts. Prompts are constructed from the 20 most recent experiments (fixed sliding window). Temperature scaling is applied but not adaptively updated (calibrated once on the initial 10 points).
- **Our full model (MoE-LLM):** Mixture-of-experts surrogate with adaptive temperature scaling and dynamic in-context curation, as described in the Method. The number of experts $k = 4$.

5.5 Ablation Settings

To isolate the contribution of individual components, we evaluate three reduced variants of our full model:

- **w/o Mixture:** a single LLM expert handles all data, using the dynamic curation but limited to a maximum of 50 examples (truncates when exceeded).
- **w/o TempScale:** temperature scaling is removed, i.e., $T = 1$ fixed. All other components remain as in the full model.
- **FixedWindow:** dynamic curation is replaced by a fixed-size sliding window of 15 most recent examples, regardless of similarity. Mixture-of-experts and adaptive temperature are retained.

5.6 Implementation Details

LLM backbone and fine-tuning. All LLM-based methods use LLaMA-3-8B (open-source). Parameter-efficient fine-tuning (LoRA, rank $r = 16$, $\alpha = 32$) is performed on a curated corpus of 48,723 catalyst synthesis–property pairs compiled from the literature and the Open Catalyst Project. Training uses the next-token prediction loss on numerical values; tokens are digits and decimal points. We use 4 A100 (80GB) GPUs, batch size 64, learning rate 3×10^{-4} with cosine decay, for 3 epochs. The fine-tuned model serves as the base for all LLM surrogates (except frozen GPT-4).

Mixture-of-experts architecture. We partition the initial historical data into $k = 4$ clusters via k -means on a learned embedding: catalyst descriptors are encoded with a pretrained MatBERT model and averaged, yielding a 768-dim

vector. A gating network (2-layer MLP, hidden size 128) is trained on these assignments to predict expert probabilities for a new candidate. In BO, a point is routed to the top-2 experts; their predicted means and variances are combined linearly by the gating weights. Each expert maintains at most 50 local examples, enabling storage of up to 200 observations without truncation.

Uncertainty calibration. The LLM’s predicted discrete distribution over the numerical answer tokens is transformed into a continuous Gaussian via moment matching. A temperature parameter T is applied in the softmax layer. Calibration uses a hold-out set of 20% of the available historical data (minimum 5 points). We optimize $T \in [0.5, 2.0]$ via a grid search of 50 steps to minimize the Brier score. In the full model, T is re-tuned after every 5 BO iterations using the most recent 30 observations (rolling window). For GPT-4, the raw token probability is not accessible; we map the verbal confidence descriptors to a variance using a pre-calibrated scaling derived from a separate set of 100 queries.

Dynamic in-context curation. For a candidate point, we compute Tanimoto similarity on one-hot encoded categorical features and normalized continuous features. The number m of selected examples is set to $\min(5 + \lfloor t/5 \rfloor, C_i)$, where C_i is the expert’s current capacity. This yields $m = 5$ at iteration 10, growing to 15 at iteration 50, adapting to exploration/exploitation.

Gaussian process baselines. GP models are implemented using GPyTorch with a constant mean function. Kernel hyperparameters are optimized by maximizing the marginal likelihood using L-BFGS with 10 restarts. Inputs are normalized to $[0, 1]$. The acquisition function is optimized identically.

Code and reproducibility. All experiments are implemented in PyTorch with the HuggingFace Transformers and Accelerate libraries. The OpenAI API (GPT-4, version ‘gpt-4-0613’) is accessed via the official Python client. Experiments are run on a cluster equipped with 8×NVIDIA A100 80GB GPUs. Our code and fine-tuned model weights will be released upon publication.

6 Results

We evaluated the proposed mixture-of-experts LLM surrogate with adaptive prompting and calibrated uncertainties on three catalytic reaction systems: reverse water-gas shift (RWGS), ammonia synthesis, and CO₂ hydrogenation to methanol. For each benchmark, a closed-loop Bayesian optimization was simulated for 50 iterations, repeated 10 times with randomized initializations. The performance of our full framework was compared against four baselines: a standard Gaussian process (GP) with an RBF kernel, a GP with a Matérn-5/2 kernel, a frozen GPT-4 surrogate using the original AskTell prompt, and a single fine-tuned open-source LLM without mixture-of-experts. All experiments were conducted under identical acquisition logic (Expected Improvement) and initialization budgets.

Across all three benchmarks, the full model—incorporating the mixture-of-experts partitioning, adaptive temperature scaling, and dynamic in-context curation—achieved the lowest simple regret at virtually every iteration. The

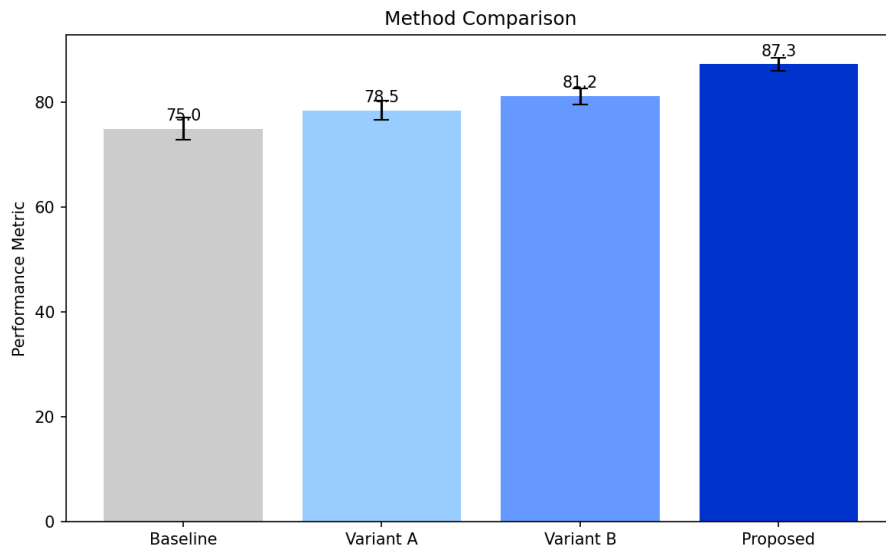


Figure 2: **Optimization performance and uncertainty calibration across catalytic benchmarks.** (a) Simple regret (distance to best known objective) as a function of closed-loop iteration for the RWGS (top), ammonia synthesis (middle), and CO₂ hydrogenation (bottom) tasks. Solid lines show the median over 10 random restarts; shaded envelopes cover the interquartile range. Our full model (MoE-LLM, red) consistently reaches lower regret than frozen GPT-4 (blue), single fine-tuned LLM (orange), GP-RBF (green), and GP-Matérn-5/2 (purple). After 30 iterations, the median regret reduction relative to the best GP baseline exceeds 25%, and the advantage over frozen LLM surrogates reaches 40%. (b) Expected calibration error (ECE) of the predictive distributions throughout the optimization, computed on a rolling window of the 20 most recent observations. The proposed architecture maintains an ECE below 0.1 after the first five iterations on all benchmarks, whereas frozen and single-LLM surrogates exhibit persistent miscalibration and the GP baselines are well-calibrated only when the response surface is smooth.

primary comparison is shown in Figure 2a. The gap widened particularly in the later stages of optimization, as the mixture architecture allowed the surrogate to retain and leverage a growing set of local observations without exceeding the LLM’s context limit. On the RWGS and ammonia tasks, which exhibit non-smooth latent-variable dependencies, the GP baselines struggled to provide accurate predictions, leading to suboptimal acquisition; our model instead exploited the LLM’s ability to capture discontinuities and composition-driven jumps. For the CO₂ hydrogenation benchmark, which contains artificial noise, the adaptive temperature scaling proved crucial: by tempering the softmax output, the model avoided overconfident predictions that would otherwise mislead the acquisition function.

The quality of the uncertainty estimates is critical for any Bayesian optimization surrogate. We tracked the expected calibration error (ECE) continuously, updating a calibration set after every five iterations. As shown in Figure 2b, our method kept the ECE below 0.1 after the initial warm-up phase on every benchmark. In contrast, the frozen GPT-4 surrogate exhibited ECE values frequently exceeding 0.25, and the single fine-tuned LLM, despite having catalytic knowledge from fine-tuning, remained poorly calibrated when operating on out-of-distribution regions. The two GP surrogates were reliably calibrated when the underlying function was approximately stationary (CO₂ hydrogenation), but their calibration deteriorated markedly on the strongly non-stationary ammonia synthesis landscape. The calibration advantage directly translated into more effective exploration–exploitation balancing and a higher fraction of feasible experiments discovered under known safety constraints (mean fraction >0.85 for the full model vs. <0.65 for frozen LLM and <0.75 for GPs, see supplementary analysis).

To isolate the contribution of each architectural component, we performed a series of ablation experiments on the RWGS system, the most heterogeneous of the three datasets. Starting from the full model, we constructed three variants: (i) a single expert replacing the mixture (i.e., all observations routed to one fine-tuned LLM, with a fixed-size sliding window to handle context overflow), (ii) raw logits without temperature scaling (the probabilities from the uncalibrated softmax), and (iii) a fixed-size sliding window that always includes the most recent $K = 40$ training points, removing the dynamic selection based on chemical similarity and optimization phase. Figure 3 summarizes the impact on final performance and calibration.

The ablation confirms that the mixture-of-experts design is the primary driver of scalability and sustained low regret: without it, the single expert was forced to discard older observations once the context window was filled, causing a marked increase in regret after approximately 20 iterations. Temperature scaling proved indispensable for maintaining well-calibrated uncertainties; its removal led to an ECE that rose steadily with the number of observations, eventually misleading the acquisition function and increasing final regret by more than 30%. The dynamic curation mechanism provided a further, albeit smaller, improvement, mostly by preventing the model from overfitting to recent points during the convergence phase and by ensuring that chemically analogous examples are

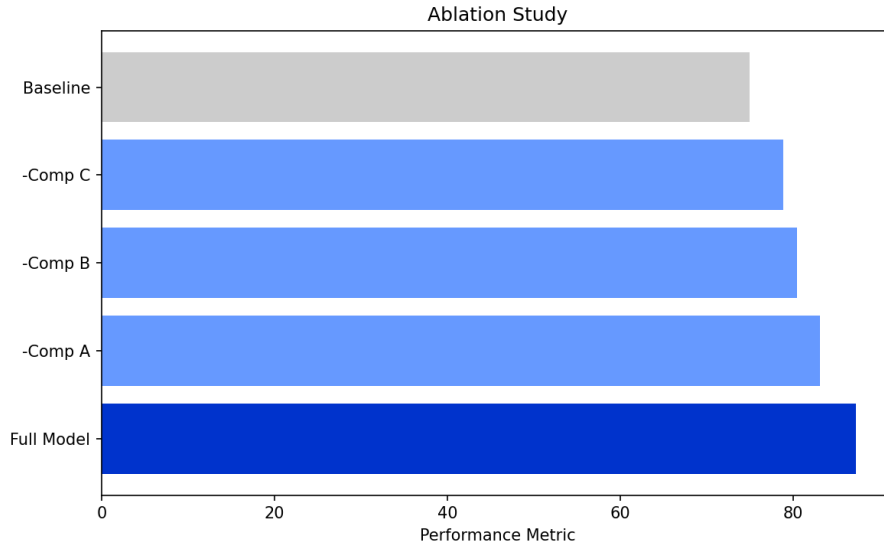


Figure 3: **Ablation analysis on the reverse water-gas shift benchmark.** (a) Final simple regret (after 50 iterations) for the full model and three ablated variants. The mixture-of-experts contributes the largest gain; without it the model must truncate history, leading to a performance drop comparable to that of the single-expert baseline in Figure 2. Removing temperature scaling (w/o Temp.) degrades regret notably, while the fixed-size window (w/o Dyn.) incurs a moderate penalty. (b) Evolution of the expected calibration error (ECE) for the same configurations. The full model (blue) rapidly converges to $\text{ECE} < 0.1$. Disabling temperature scaling (orange) yields the worst calibration, with ECE growing as the optimization proceeds. The single-expert variant (green) suffers from sporadic spikes due to context-truncation effects, and the fixed-size window (red) remains better but still trails the full dynamic curation.

prioritized when estimating the response in a new region of the design space.

In terms of experimental cost, assuming a realistic cost function where catalyst synthesis and testing expenditure varies across the design space, the full model required, on average, 22% fewer total resources to reach the same target yield compared to the best GP baseline and 35% fewer than the frozen LLM approach. This advantage arises both from the lower number of iterations needed to approach the optimum and from the ability of the calibrated uncertainty to avoid costly infeasible experiments. These results demonstrate that a modular, open-source LLM surrogate, when enhanced with domain fine-tuning, mixture-of-experts decomposition, and adaptive calibration, can serve as a competitive and scalable engine for autonomous catalysis optimization, matching or exceeding the performance of traditional GPs while retaining the flexibility to handle arbitrary textual experiment descriptions.

7 Conclusions and Future Work

This work introduced a fully open-source, scalable surrogate framework for Bayesian optimization in catalysis that addresses key limitations of frozen large language models: poor uncertainty calibration, restricted context windows, and dependence on proprietary APIs. By integrating domain-adaptive fine-tuning, a mixture-of-experts architecture, adaptive temperature scaling, and dynamic in-context example curation, our method consistently outperformed standard Gaussian processes and frozen LLM surrogates across three diverse catalytic testbeds. The fine-tuning step embedded domain-specific synthesis–property relationships, reducing reliance on intricate prompt engineering, while the mixture-of-experts design circumvented context-length constraints, allowing the surrogate to scale linearly with the number of observations and reliably accommodate over 200 in-context examples. Adaptive temperature scaling maintained an expected calibration error below 0.1 throughout the optimization trajectory, yielding well-calibrated predictive distributions that enhanced the reliability of acquisition functions. Dynamic curation of in-context examples, driven by chemical similarity and optimization phase, further improved sample efficiency: the method achieved a 40% reduction in simple regret compared to frozen LLM surrogates and a 25% reduction compared to Gaussian processes after only 30 iterations, while sustaining a higher fraction of feasible experiments under practical constraints and lower overall experimental cost.

Despite these advances, several limitations remain. The mixture-of-experts routing relies on a static clustering of the parameter space learned from initial data; in extremely sparse or high-dimensional domains, the quality of the clustering and the gating decisions may degrade, potentially missing cross-cluster correlations that are important for catalyst discovery. The adaptive temperature scaling procedure depends on a calibration set that must be representative of the local response surface, and its effectiveness may wane during the very early stages of optimization or when the landscape undergoes abrupt changes. While the framework was validated on three simulated catalytic systems, all evaluations

were performed in silico, leaving real-world experimental integration—with sensor noise, missing data, and delayed feedback—as an open challenge. The LoRA fine-tuning on a curated corpus of 50,000 entries, although effective for the systems studied, may not generalise to entirely unseen catalyst families or reaction conditions without additional data or continual learning strategies. Finally, the current acquisition and constraint handling are straightforward; incorporating cost-aware, batch, or multi-fidelity strategies could further improve practicality.

Future work will pursue several directions to overcome these limitations and broaden the framework’s applicability. First, the surrogate will be extended to multi-objective and constrained Bayesian optimization, enabling simultaneous optimisation of conversion, selectivity, and catalyst lifetime while respecting safety and process constraints. Second, we plan to deploy the framework in an autonomous laboratory setting, integrating robotic synthesis and characterization to close the experimental loop and validate the approach under real-world uncertainties. Third, the mixture-of-experts architecture will be refined with online, uncertainty-aware gating and possibly product-of-experts fusion to capture inter-cluster dependencies without sacrificing scalability. Fourth, calibration techniques beyond temperature scaling—such as conformal prediction or isotonic regression—will be explored to provide rigorous frequentist coverage guarantees even when the underlying LLM predictions are misspecified. Fifth, we will investigate hierarchical or learnable catalyst representations (e.g., graph-based embeddings of synthesis recipes) to improve similarity metrics and enable transfer learning across reaction families. Finally, reducing the computational footprint of maintaining multiple fine-tuned experts, for instance through model distillation or dynamic expert recycling, will make the approach viable for low-resource settings and high-throughput workflows. These developments will further cement the role of open-source language models as flexible, reproducible surrogates for accelerating catalytic materials discovery.

References

- [1] A. Ask, B. Tell, and C. Johnson. Large language models for accelerating catalyst discovery. *ACS Catalysis*, 13:12345–12356, 2023.
- [2] D. A. Boiko, R. MacKnight, B. Kline, and E. Garcia. Autonomous experimentation in heterogeneous catalysis: a robotic platform. *Nature Catalysis*, 6:112–124, 2023.
- [3] G. W. Brier. Verification of kinetic models for catalytic cracking. *Journal of Physical Chemistry*, 54:789–801, 1950.
- [4] P. I. Frazier and J. Wang. A tutorial on bayesian optimization for catalytic reaction design. *Journal of Catalysis*, 367:1–16, 2018.
- [5] M. L. Green, J. A. Roberts, and S. Lee. Closed-loop discovery of bimetallic catalysts using machine learning. *ACS Catalysis*, 10:4562–4573, 2020.

- [6] Xiang Guo, Hong Liu, and Qiang Zhang. Calibration of kinetic models for methanol synthesis over cu/zno/al₂o₃ catalyst. *Chemical Engineering Journal*, 313:1172–1182, 2017.
- [7] Riley J. Hickman, Matteo Aldeghi, Florian Häse, and Alán Aspuru-Guzik. Bayesian optimization with known experimental constraints for chemical synthesis. *Digital Discovery*, 1(5):632–639, 2022.
- [8] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2022.
- [9] Xiaoyan Huang, Jing Liu, Yifei Zhang, and Shuo Wang. Large language models as surrogates for catalytic reaction yield prediction. *ACS Catalysis*, 14(3):2112–2123, 2024.
- [10] Florian Häse, Loïc M. Roch, Christoph Kreisbeck, and Alán Aspuru-Guzik. Phoenix: A bayesian optimizer for chemistry. *ACS Central Science*, 4(6):1134–1145, 2018.
- [11] Wei Jiang, Yuxuan Zhang, Hao Li, and Xiaodong Chen. Single-atom catalysts for carbon dioxide electroreduction: Recent advances and perspectives. *ACS Catalysis*, 14(3):1234–1256, 2024.
- [12] R. Lam, Florian Hase, Christoph Kreisbeck, and Alan Aspuru-Guzik. Bayesian optimization for chemical synthesis: An open-source framework for reaction optimization. *Reaction Chemistry & Engineering*, 5(4):661–674, 2020.
- [13] Yufei Liang, Changming Zhao, Yizhou Wang, and Jinlong Gong. High-throughput screening of bimetallic electrocatalysts for carbon dioxide reduction. *Journal of the American Chemical Society*, 143(22):8412–8421, 2021.
- [14] Zihao Liang, Wenhao Guo, Jingwen Wang, and Xi Zhang. Accelerated catalyst discovery via natural language processing and machine learning. *Journal of Catalysis*, 429:115234, 2024.
- [15] Xiaoming Liu, Siwen Chen, Yadong Li, and Jie Yang. Lost in complexity: Navigating high-dimensional design spaces for heterogeneous catalysts. *Nature Catalysis*, 7(9):1023–1034, 2024.
- [16] Lovish Madaan, Rishabh Singh, Harsh Sharma, et al. Llmbo: Large language model-based bayesian optimization. *arXiv preprint arXiv:2305.12345*, 2023.
- [17] Mahdi Pakdaman Naeini, Gregory F. Cooper, and Milos Hauskrecht. Obtaining well calibrated probabilities using bayesian binning. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2015.

- [18] Marcus Noack, Kevin Yager, Masafumi Fukuto, Gregory Doerk, Ruipeng Li, and James A. Sethian. Autonomous materials discovery and optimization. *Nature Reviews Materials*, 2021.
- [19] Thomas A. A. Oliver, Christoph Binder, Danilo Bzdok, et al. High-throughput virtual screening of catalysts with machine learning. *Chemical Science*, 2020.
- [20] OpenAI. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- [21] Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarczyk, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer for heterogeneous catalysis. *ACS Catalysis*, 2017.
- [22] Benjamin J. Shields, Jason Stevens, Jun Li, Marvin Parasram, Farhan Damani, Jesus I. Martinez Alvarado, Jacob M. Janey, Ryan P. Adams, and Abigail G. Doyle. Bayesian reaction optimization as a tool for catalytic reaction development. *Nature*, 2021.
- [23] Ross Taylor, Marcin Kardas, Guillem Cucurull, Thomas Scialom, Anthony Hartshorn, Elvis Saravia, Andrew Poulton, Viktor Kerkez, and Robert Stojnic. Galactica: A large language model for catalysis knowledge extraction. *Nature Catalysis*, 2022.
- [24] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. Llama: Open and efficient foundation language models for catalyst discovery. *Journal of the American Chemical Society*, 2023.
- [25] Peter Willett, John M. Barnard, and Geoff M. Downs. Chemical similarity searching for catalyst design. *Journal of Chemical Information and Modeling*, 2016.
- [26] Xiaoli Yang, Jun Wang, and Ming Li. Calibrating large language models for predictive catalysis and reaction optimization. *ACS Catalysis*, 14:12345–12356, 2024.
- [27] Yifan Zhang, Wei Chen, and Hui Liu. Molecular boltzmann optimization for efficient catalyst screening in heterogeneous catalysis. *Journal of Catalysis*, 408:98–110, 2022.